

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2023-24

Paper: BVCA-303 — Programming In Linux

Time: Three hours

Maximum Marks: 70

Instructions: Answer the questions in each section as instructed. All questions carry marks as indicated.

SECTION — A

Long-Answer Questions. Answer any two questions:

2 × 12 = 24 Marks

Question 1. Explain the architecture of Linux, emphasizing the roles of the kernel, shell, and file system.

Question 2. Describe the Linux file system hierarchy in detail, providing examples of directories at each level. Explain the purpose and usage of essential file system management commands such as `ls`, `cp`, `mv`, and `rm`. Illustrate their functionality with practical examples.

Question 3. Discuss the significance of Linux commands in performing various tasks efficiently. Explain how commands like `touch`, `grep`, `wc`, and `cat` contribute to file manipulation, searching, and text processing. Provide examples to demonstrate the practical utility of these commands in everyday computing tasks.

Question 4. Compare and contrast the advantages and disadvantages of using Linux as an operating system. Provide real-world examples to support your arguments.

SECTION — B

Question 5. Attempt any six parts (Answer in about 200 words each): **6 × 6 = 36 Marks**

(a) Write a shell script to print a pattern of asterisks (*) in a triangular format:

```
*  
**  
***  
****  
*****
```

(b) Develop a shell script that accepts two numbers as input and performs addition, subtraction, multiplication, and division on them.

(c) Discuss the significance of pipes in Linux. Provide an example scenario where pipes are used effectively with multiple commands.

(d) Discuss the purpose of the `mkdir` command in Linux. Provide an example demonstrating its usage in creating directories.

- (e) Define the **tee** command in Linux and explain its functionality.
- (f) What is the purpose of the **cat** command in Linux? How does it differ from other file viewing commands?
- (g) Explain the significance of the **banner** command in Linux. How can it be used to display text in a visually appealing manner?
- (h) Discuss the functionality of the **grep** command in Linux.
- (i) Describe the significance of the **cal** command in Linux. How does it differ from other calendar-related commands? Provide an example demonstrating its usage.

SECTION — C

Question 6. Attempt all parts:

10 × 1 = 10 Marks

- (a) Which command is used to display the current working directory in Linux?
 - (a) **ls**
 - (b) **pwd**
 - (c) **cal**
 - (d) **cp**
- (b) Which command is used to create a new directory in Linux?
 - (a) **mkdir**
 - (b) **touch**
 - (c) **rm**
 - (d) **mv**
- (c) What does the **man** command do in Linux?
 - (a) Displays the manual page of a command
 - (b) Moves files or directories
 - (c) Creates a new file
 - (d) Counts the number of words in a file
- (d) What does the **head** command do in Linux?
 - (a) Displays the first part of a file
 - (b) Displays the last part of a file
 - (c) Searches for a pattern in a file
 - (d) Counts the number of lines in a file
- (e) Which command is used to count lines in a file in Linux?

- (a) `wc`
 - (b) `sort`
 - (c) `grep`
 - (d) `banner`
- (f) What is the purpose of the `grep` command in Linux?
- (a) Displays the calendar for the current month
 - (b) Displays the current date and time
 - (c) Counts the number of lines in a file
 - (d) Searches for a pattern in a file
- (g) Which command is used to count the number of words in a file in Linux?
- (a) `mv`
 - (b) `wc`
 - (c) `ps`
 - (d) `tee`
- (h) What is the function of the `sort` command in Linux?
- (a) Displays the sorted content of a file
 - (b) Moves files or directories
 - (c) Counts the number of lines in a file
 - (d) Searches for a pattern in a file
- (i) Which command is used to display the last part of a file in Linux?
- (a) `head`
 - (b) `tail`
 - (c) `who`
 - (d) `bc`
- (j) What does the `mkdir` command do in Linux?
- (a) Removes a directory
 - (b) Creates a new directory
 - (c) Moves files or directories
 - (d) Displays the manual page of a command

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